Jiaju Wu

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Education

- **Sun Yat-sen University**
  M.S. in Physics (Optics), School of Science (Recommended Admission) Sep 2025 – Present
  - Grade: 91/100, supervisor: Assoc. Prof. Xiantao Jiang

  B.S. in Physics, School of Physics and Astronomy Sep 2021 – Jun 2025
  - GPA: 3.6/4.0, Rank: 28/113
  - Outstanding Bachelor's Thesis (Top 7%)

Research Experience

Research areas: Structured light, laser speckle, deep learning, scattering imaging and quantitative phase microscopy

- **SYSU | Optics Imaging Lab** Feb 2025 – Present
  - **Project 1: *Dual-dimensional Multiplexing Optical Communication through Scattering Media with Generalized Perfect Optical Vortex***
    - Proposed a GPOV-based geometry-OAM joint encoding scheme for scattering-resistant optical communication transmission
    - Proposed a correlation-guided residual network (CG-ResNet) with GPOV speckle correlation prior for dual-dimensional recognition; achieved 97.08% average OAM accuracy and 98.75% geometry accuracy under unknown scattering conditions
    - Realized grayscale and RGB image transmission through strongly scattering media with bit error rate (BER) below 1%
  - **Project 2: *High-Order Orbital Angular Momentum Extraction from Vortex-Beam Speckles with Low Spatial Sampling***
    - Revealed the physical mechanism behind speckle correlation degradation for high-order OAM modes
    - Proposed a low-complexity topological charge-matching strategy to extend the accessible range of high-order OAM retrieval
    - Realized accurate OAM extraction with only 0.3% speckle pattern maintaining robustness against strong scattering
  - **Project 3: *Simulation and Measurement of the Point Spread Function of Laser Speckle Patterns*** (B.S. Thesis, Outstanding)
    - Conducted numerical simulations on the formation and propagation of laser speckles, and experimentally measure the speckle point spread function (PSF) and the range of optical memory effect (OME)
    - Realized speckle autocorrelation imaging based on phase retrieval algorithms (GS, HIO-ER); achieved speckle PSF deconvolution imaging by adopting Lucy-Richardson and Wiener filtering algorithms

  Supervisor: Assoc. Prof. Xiantao Jiang

- **SYSU | MOE Key Laboratory of TianQin Mission** Nov 2023 – Oct 2024
  - **Project 4: *Theoretical Modeling and Numerical Simulation of Multi-Beam Interference Based on the Astigmatic Gaussian Beam Model*** (Undergraduate Innovation Training Program, Project Leader)
    - Proposed an astigmatic Gaussian beam simulation scheme for TianQin laser interferometer, quantifying multi-beam interference effects and stray light noise in space gravitational wave detection
    - Quantitatively evaluated the stray light noise suppression effect of balanced detection in picometer-sensitivity dual-frequency heterodyne laser interferometer

  Supervisor: Assoc. Prof. Huizong Duan

Publications

The first-author:

- ***Jiaju Wu***, Rui Cao, Shengqiang Zhong, Yuhan Liu, Kaibin Zeng, Rushi Yang, Li Yang, Xiantao Jiang. "High-Order Orbital Angular Momentum Extraction from Vortex-Beam Speckles with Low Spatial Sampling," *Under review at Optics & Laser Technology, 2026.*
- ***Jiaju Wu***, Yuhan Liu, Rushi Yang, Wenkai Wang, Kaibin Zeng, Shengqiang Zhong, Han Zhang, Xiantao Jiang. "Dual-Dimensional Multiplexing Optical Communication through Scattering Media with Generalized Perfect Optical Vortex,"*Submitted for publication, 2026.*

Other:

- Kaibin Zeng, Weiyuan Li, Wenjun Zhang, Bin Liu, ***Jiaju Wu***, Shengqiang Zhong, Hongwei Zou, Fan Yang, Chao Hou, Lei Xu, Xianglong Yu, Xin Luo, Xiantao Jiang. "Precise Characterization of Optical Constant of Two Dimensional WS2 Using Diffraction Phase Microscopy," *Optics & Laser Technology* 192, 114109 (2025). (JCR Q1, IF=5.2)
- Kaibin Zeng, ***Jiaju Wu***, Xiantao Jiang. "A General Phase Denoising Method for High-Sensitivity Diffraction Phase Imaging," *Submitted for publication, 2026.*
- Yuhan Liu, Hongwei Zou, Shengqiang Zhong, ***Jiaju Wu***, Kaibin Zeng, Rushi Yang, Xiantao Jiang. "Structured Light-Modulated

Speckle Field for Three-Dimensional High-Resolution Biomedical Imaging," *Submitted for publication, 2026.*

- Shengqiang Zhong, Chunyan Tao, Yuhan Liu, ***Jiaju Wu***,Hongwei Zou, Kaibin Zeng, Rushi Yang, Yuanzhi Shao, Yongsheng Huang, Xiantao Jiang. "High-Resolution Three-Dimensional Refractive Index Tomography via Vortex Beam Speckle Illumination," *Submitted for publication, 2026.*

Honors & Awards

- **First-class Scholarship** of Sun Yat-sen University 2025
- **Meritorious Winner** (Top 7%) in The Interdisciplinary Contest in Modeling (ICM) 2024
- **Second-class Scholarship** of Sun Yat-sen University 2023
- **Project Leader** of Undergraduate Innovation Training Program Grant (Funded by SYSU) 2023
- **Third-class Scholarship** of Sun Yat-sen University 2023
- **Secondary Prize** of The Mathematics competition of Chinese College Students 2022
- **Secondary Prize** of The Asia and Pacific Mathematical Contest in Modeling 2022
- **Special Scholarship** of Sun Yat-sen University (Outstanding Conduct Award) 2022
- **First Prize** (Team Leader) of China Undergraduate Physics Tournament (CUPT, SYSU Selection) 2021

Skills

Languages: English (CET-6: 566), Mandarin Chinese (native)
Programming and Tools: MATLAB, Python (PyTorch, scikit-learn...), C++, Lumerical FDTD, COMSOL, Origin, LaTeX, Blender

Teaching / Mentorship

Teaching Assistant: *Probability and Statistics for Science and Engineering* (Autumn 2025)
Mentorship: Mentored an undergraduate research project on *Speckle Formation Mechanisms Using FDTD Simulations*.